Preliminary Results of Gaze Tracking and Hand Motor Synergy Coordination During Reach-to-Grasp VR Tasks for Improved Prosthetic Arm Intent Prediction

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Current prosthetic limbs require subjects to actively control individual joints sequentially, which can lead to awkward compensatory motions and cognitive burden. In this work we present initial results from our exploration of gaze and motion synergies during arm movements. The long-term goal for this work is to understand whether by tracking a person's gaze point position and acceleration we can make inferences regarding the motion of their hand, towards the development of a naturalistic prosthesis controller.

Specifically, we analyzed gaze and motion-capture (MoCap) data from the open-access 3D-ARM-Gaze dataset[1], in which able-bodied subjects performed reaching tasks in a virtual reality (VR) environment. We sought to quantify timing of the gaze and hand motion, to determine whether an individual's gaze focus-point will precede the motion of their hand, and to characterize relevant gaze and hand kinematic behaviors during reaching tasks. Our methods involved transforming the dataset's coordinates by rotating the data into a frame of reference with the horizontal axis (or "depth") aligned with the subjects' line of sight at the beginning of each reach. This is because we believe that subjects may plan their reaches in this rotated reference frame to enable better hand-eye coordination.

Our preliminary results show that the peak acceleration of the gaze focus point has a consistent lead-time of approximately $150\ ms$, relative to the hand's peak acceleration during the initial stages of a reaching motion. While this striking difference alone suggests that the intention to move the hand can be inferred relatively early-on from a person's gaze, the current results are not exact enough to allow gaze-point to be the only signal used for intention or end-point predictions. Despite expectations, our results failed to show coupling between the location of the gaze focus point and the location of the hand – after transforming the reference frame into "eye-hand" coordinates, for some subjects the gaze was inversely related along the two orthogonal axes, while for other subjects the gaze trajectory in the y- and z- axes was parallel. Meanwhile, the hand did appear to move relatively smoothly within the rotated "eye-hand" reference frame, which may lend support to previous literature showing that we coordinate end-point behavior during arm movements in eye or hand coordinates. We will further explore this in our future work.

Since the 3D-ARM-Gaze dataset consists of location, orientation and time data for the gaze and upper-body segments during non-naturalistic reach-and-grasp tasks, our next major step will be building a custom dataset. This new multi-modal dataset will consist of gaze tracking, surface electromyography (sEMG) and kinematic data that will be recorded during real-world naturalistic activities of daily living (ADLs). We intend to use this new dataset to better

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[1] Lento et al., "Data and Code for 3D-ARM-Gaze."

understand the spatiotemporal coordination between muscle, postural, and gaze synergies that exists in upper-limb movements, and we aim to eventually leverage that knowledge to predict user intent and develop a prosthetic arm controller that enables naturalistic motion with minimal active input.